

120+ COMMANDS

Raspberry Pi Commands Cheat Sheet

The essential terminal commands for Raspberry Pi OS – from first steps through GPIO and the camera to backup & bootloader.

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Whether you just flashed a fresh SD card or you've been running your Pi for years, sooner or later you end up staring at the terminal, unable to remember the exact command. This **Raspberry Pi commands cheat sheet** collects over 120 terminal commands for Raspberry Pi OS in one place – from first steps through GPIO and the camera to bootloader updates and SD card backups. Not a tutorial to read through, but a reference to look up and copy.

> **What is a Raspberry Pi cheat sheet?**
>
> A Raspberry Pi cheat sheet is a compact reference list of the most important terminal commands for Raspberry Pi OS, grouped by topic such as system, network, GPIO, or backup. It's not a substitute for a tutorial — it's a quick lookup for commands you don't want to memorize permanently.

> **How to use this cheat sheet:** Every command is grouped by topic and meant to be copied directly. Replace placeholders like `<path>` or `<service>` with your own values. Commands prefixed with `sudo` need admin rights.

> **Prefer a printable version?**
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First Steps & System Basics

The basics you need first on any freshly set-up Pi.

Command	What it does
`sudo raspi-config`	Opens the central menu-driven configuration tool: locale, boot behavior, interfaces (SSH, I2C, SPI, camera), overclocking – most base settings live here.
`passwd`	Changes the password of the currently logged-in user – asks for the old password first, then the new one twice.
`sudo reboot`	Restarts the Pi cleanly, shutting down all running services properly first.
`sudo shutdown -h now`	Shuts the Pi down safely right away (h = halt) – important before pulling the power cable, or you risk SD card corruption.
`sudo shutdown -r +5`	Warns all logged-in users about a restart in 5 minutes, then reboots automatically.
`whoami`	Prints the username of the currently active terminal user – handy after an `su` or `sudo -i`.
`hostname`	Shows just the device name (no domain) the Pi announces on the network.
`hostname -I`	Lists every IPv4/IPv6 address assigned to the Pi (LAN, Wi-Fi, VPN) on one line – see also our full guide to finding the IP address.

Command	What it does
`uname -a`	Prints kernel name, version, build date, and architecture (e.g. aarch64) in one line – useful for compatibility checks.
`cat /etc/os-release`	Shows the name, version number, and codename of the installed Raspberry Pi OS – more reliable than `uname` for the OS version.
`date`	Shows the current date and time including timezone, e.g. to cross-check cron or log timestamps.
`timedatectl`	Shows local time, UTC, timezone, and NTP sync status; use `sudo timedatectl set-timezone` to change the timezone directly.
`uptime`	Shows time since the last boot plus system load over the last 1/5/15 minutes (load average).

System Info & Diagnostics

`vcgencmd` is Pi-exclusive (Broadcom firmware interface) and doesn't work on plain Debian.

Command	What it does
`vcgencmd measure_temp`	Reads the SoC temperature straight from the sensor, e.g. `temp=42.8'C` – above 80°C the Pi throttles the clock, above 85°C it's throttled further.
`vcgencmd get_throttled`	Checks whether the Pi has been throttled due to low power or high temperature. `0x0` means everything's fine – any other value means this has happened, either right now or at some point since the last boot.
`vcgencmd measure_volts`	Shows the voltage of the `core` component by default; use `sdram_c` / `sdram_i` / `sdram_p` for the RAM areas.
`vcgencmd measure_clock arm`	Shows the current CPU clock frequency in Hz – compare it against the overclocking value set in raspi-config.
`vcgencmd get_config int`	Lists every integer config.txt parameter (e.g. `arm_freq`, `over_voltage`) with its currently active value.
`vcgencmd version`	Shows the version number and build date of the GPU firmware/bootloader combo – useful when reporting bugs.
`free -h`	Shows RAM and swap usage in readable units (h = MB/GB instead of bytes), including available and cached memory.
`df -h`	Shows used/free space on all mounted filesystems in readable units – useful for catching a full SD card early.
`du -sh <folder>`	Calculates the total size of a folder (s = summarized, h = readable) – great for tracking down space hogs before cleaning up.
`top`	Shows running processes live, sorted by CPU usage – press `q` to quit, `P` / `M` to re-sort by CPU or memory.

Command	What it does
`htop`	Colorful, scrollable, mouse-friendly alternative to `top` with a per-core usage bar (install with `sudo apt install htop` if needed).
`lscpu`	Shows core count, architecture, clock range, and cache sizes of the CPU in a structured format.

> **Practical tip:** For 24/7 operation and backups (see section 10), it's worth using a card with high write endurance, e.g. the Samsung Pro Endurance microSD – regular consumer cards aren't built for round-the-clock logging. More in our SD card comparison.

Files & Directories

Command	What it does
`ls -la`	Lists all files including hidden ones (`-a` , starting with a dot) in long format (`-l` : permissions, owner, size, date).
`cd <path>`	Changes the working directory; `cd ..` goes up one level, `cd` with no argument returns to the home directory.
`pwd`	Prints the full, absolute path of the current directory (print working directory).
`cp <source> <target>`	Copies a single file – an existing target is overwritten without confirmation.
`cp -r <source> <target>`	Copies an entire folder with everything inside it, including nested subfolders (that's what the extra `-r` flag does) – without it, `cp` fails with an error on folders.
`mv <source> <target>`	Moves a file/folder, or renames it if source and target are in the same directory.
`rm <file>`	Deletes a file permanently – there's no trash bin and no default undo.
`rm -r <folder>`	Deletes a folder and everything inside it, including nested subfolders (`-r`) – combined with `-f` (force) there's no confirmation at all, so use with extra caution.
`mkdir <name>`	Creates a new, empty folder; `-p` also creates any missing parent folders.
`touch <file>`	Creates an empty file if it doesn't exist, or just updates the timestamp of an existing one.
`nano <file>`	Opens the file in the beginner-friendly terminal editor – save with `Ctrl+O` , exit with `Ctrl+X` .

Command	What it does
`cat <file>`	Prints the entire file content at once – unwieldy for long files, where `less` is the better choice.
`less <file>`	Opens the file for page-by-page browsing (arrow keys, `/` to search) without loading it entirely into memory.
`find <path> -name "<pattern>"`	Searches a folder and all of its subfolders for filenames; the pattern supports wildcards like `*.log` for "anything ending in .log".
`grep -r "<text>" <path>`	Searches a folder and all of its subfolders (`-r`) for a piece of text and shows the filename plus the matching line for every hit.
`tar -czvf archive.tar.gz <folder>`	Packs a folder into a compressed archive (`c` = create, `z` = gzip, `v` = verbose, `f` = filename).
`tar -xzvf archive.tar.gz`	Extracts a gzip-compressed tar archive (`x` = extract) into the current directory.
`chmod +x <file>`	Sets the execute permission for owner, group, and others – required so scripts can be run directly with `./script.sh` .
`chown pi:pi <file>`	Changes the owner (`pi`) and group (`pi`) of a file – often needed when files were created as root, e.g. via FTP.

Network & SSH

Command	What it does
`ip a`	Shows all network interfaces (`eth0` , `wlan0` , `lo`) with their IPv4/IPv6 addresses and status (UP/DOWN) – more detailed than `hostname -I` .
`ping <address>`	Sends ICMP packets to a target and measures response time – stop with `Ctrl+C` , limit to 4 packets with `-c 4` .
`nmcli device wifi list`	Lists every Wi-Fi network in range with signal strength and encryption type.
`nmcli device wifi connect "<SSID>" password "<password>"`	Connects to a Wi-Fi network directly from the command line, without opening raspi-config.
`iwgetid`	Shows the SSID of the currently connected Wi-Fi network – the fastest check for whether a Wi-Fi connection exists at all.

Command	What it does
`sudo raspi-config nonint do_ssh 0`	Enables SSH non-interactively via script (0 = on, 1 = off) – ideal for automating headless setups, see also our SSH tutorial.
`ssh pi@<ip-address>`	Opens an encrypted terminal connection to the Pi – on first connect you must confirm the host fingerprint.
`scp <file> pi@<ip>:<target-path>`	Copies a file to the Pi over SSH, same syntax as `cp` with a host address prepended.
`ssh-keygen -t ed25519`	Generates a modern, secure SSH key pair – transfer the public key to the Pi afterwards with `ssh-copy-id` for passwordless login.
`curl -I <URL>`	Fetches only the HTTP response headers without downloading the page content – a quick test whether a server is reachable and what status it returns.
`wget <URL>`	Downloads a file straight into the current directory, and supports resuming interrupted downloads with `-c`.
`sudo raspi-config nonint get_hostname`	Prints the currently set hostname – the counterpart to `do_hostname` for setting a new one.

Package Management (APT)

Command	What it does
`sudo apt update`	Refreshes the package lists from the configured repositories – installs nothing yet, but is required before upgrade/install.
`sudo apt upgrade -y`	Upgrades already-installed packages to newer versions, without installing new dependencies or removing packages (`-y` auto-confirms).
`sudo apt full-upgrade -y`	Like upgrade, but may also remove or install packages if needed to resolve version conflicts – recommended for larger OS updates.
`sudo apt install <package>`	Installs a new package along with all required dependencies.
`sudo apt remove <package>`	Uninstalls the program but leaves config files in `/etc` untouched – handy if you plan to reinstall later.
`sudo apt purge <package>`	Removes the program and its config files completely – for a truly clean reinstall.

Command	What it does
`sudo apt autoremove`	Removes automatically installed dependencies that no remaining package needs anymore.
`apt list --installed`	Lists every package installed via apt/dpkg with its version – combines well with `| grep <name>`.
`apt-cache search <term>`	Searches package names and short descriptions in the repositories by keyword, before you know the exact package name.
`dpkg -l | grep <name>`	Checks directly via dpkg (the layer beneath apt) whether and in which version a package is installed.
`sudo apt clean`	Deletes downloaded .deb install files from `/var/cache/apt/archives` – frees up space quickly when storage is tight.

Processes & Services (systemd)

Command	What it does
`systemctl status <service>`	Shows whether a service is active/inactive/failed, since when, and includes the most recent log lines.
`sudo systemctl start <service>`	Starts a service right away, without changing its autostart behavior on the next boot.
`sudo systemctl stop <service>`	Stops a running service immediately, without disabling it permanently.
`sudo systemctl restart <service>`	Stops and starts a service again in one step – the standard move after a config change.
`sudo systemctl enable <service>`	Makes the service start automatically on every system boot.
`sudo systemctl disable <service>`	Removes the autostart entry – the service then needs to be started manually, though it may still be running until the next reboot.
`systemctl list-units --type=service`	Lists all currently loaded services with their status (active/inactive/failed).
`ps aux`	Shows all running processes from every user with PID, CPU/RAM usage, and the full command.
`ps aux | grep <name>`	Filters the process list by name – the fastest way to find a program's PID.
`kill <PID>`	Politely asks a process to shut down – it gets a chance to clean up first (e.g. save files), but won't always respond right away.
`kill -9 <PID>`	Terminates a process immediately, with no chance to clean up first – only use this if plain `kill` doesn't work.

Command	What it does
`killall <name>`	Terminates every process with this exact name at once, without having to look up the PID first.
`journalctl -xe`	Shows the most recent system logs with extended explanations (`-x`) jumped to the end of the log (`-e`) – the first step for any error after boot.
`journalctl -u <service> -f`	Follows a specific service's log live (`-f` = follow, like `tail -f`) – ideal for debugging while it starts.

GPIO & Hardware Interfaces

> **Tip:** You don't need to memorize the pin layout itself – our interactive **GPIO pinout tool** shows pin number, function, and board position in one click for Pi 5, 4, 3, and Zero. The commands below are for direct access from the terminal.

> **Note:** The classic `gpio readall` command from the WiringPi library is considered deprecated and is no longer included in current Raspberry Pi OS repositories. Use `raspi-gpio` for new projects instead. More background in the GPIO beginner's tutorial.

Command	What it does
`raspi-gpio get`	Shows function (mode), pull-up/down resistor, and current level (0/1) for every pin in a compact overview.
`raspi-gpio get <pin>`	Shows mode, pull resistor, and level for a single pin only, e.g. `raspi-gpio get 17` .
`raspi-gpio set <pin> op dh`	Configures the pin as output (`op`) and drives it high/3.3V directly (`dh` = drive high).
`raspi-gpio set <pin> op dl`	Configures the pin as output and drives it low/0V (`dl` = drive low).
`raspi-gpio funcs`	Lists every available alternate hardware function (e.g. I2C, SPI, UART, PWM) for each pin.
`sudo raspi-config nonint do_i2c 0`	Enables the I2C interface including the required kernel modules, without opening the menu (0 = on, 1 = off).
`sudo raspi-config nonint do_spi 0`	Enables the SPI interface non-interactively – a prerequisite for many displays and ADC modules.
`i2cdetect -y 1`	Scans I2C bus 1 (the default on current models) and shows the addresses of all detected devices in a table.

Command	What it does
`sudo raspi-config nonint do_serial_hw 0`	Enables the hardware UART (`ttyAMA0`) for reliable serial communication, e.g. with GPS modules or RTC boards.

Camera Module

> **Note:** Current Raspberry Pi OS versions use the `rpicam-*` command set (successor to `libcamera-*` and the old `raspistill`). The old command names still work in some cases because they're automatically redirected to the new ones behind the scenes, but shouldn't be used in new scripts anymore.

Command	What it does
`rpicam-hello`	Opens a live camera preview for a few seconds on the connected display – the simplest functional test.
`rpicam-still --list-cameras`	Lists every camera module detected by the system, with sensor name and supported resolutions.
`rpicam-still -o image.jpg`	Captures a single photo after a brief autofocus/exposure adjustment and saves it under the given name.
`rpicam-vid -t 10000 -o video.h264`	Records a video for the duration given in `-t` , in milliseconds (here: 10 seconds).

Bootloader, Firmware & Updates

Pi-exclusive: EEPROM bootloader management only exists on real Raspberry Pi hardware (Pi 4/5).

Command	What it does
`vcgencmd bootloader_version`	Shows the version date and release channel (critical/stable/beta) of the currently flashed EEPROM bootloader.
`sudo rpi-eeprom-update`	Compares the installed version against the latest available bootloader version, without changing anything (a pure check).
`sudo rpi-eeprom-update -a`	Installs the latest available EEPROM version automatically – written by `recovery.bin` on the next reboot.
`sudo raspi-config nonint do_expand_rootfs`	Expands the root partition to the full capacity of the SD card/SSD – already done automatically on first boot on most current images.

Command	What it does
`sudo apt update && sudo apt full-upgrade -y`	Combined command for a full system update including kernel and firmware packages in one go.

Storage, SD Card & Backup

> **Careful with `dd`:** `if=` (input) and `of=` (output) must never be swapped – a mix-up will irreversibly overwrite the wrong disk. Always check drive names first with `lsblk`. Find a detailed walkthrough in our backup tutorial.

Command	What it does
`lsblk`	Shows all block devices as a tree with partitions, sizes, and mount points – the safest first step before any `dd` command.
`sudo fdisk -l`	Shows detailed partition tables (type, start/end sector, size) for all detected drives.
`sudo mount /dev/sda1 /mnt`	Manually mounts partition sda1 into the (empty) `/mnt` directory, accessible through `/mnt` afterward.
`sudo umount /mnt`	Cleanly unmounts a previously mounted drive – essential before physically unplugging it, to avoid data loss.
`sudo dd if=/dev/sdX of=backup.img bs=4M status=progress`	Creates an exact 1:1 image of the entire card including the partition table; `bs=4M` speeds up the transfer, `status=progress` shows live progress.
`sudo dd if=backup.img of=/dev/sdX bs=4M status=progress`	Writes a previously created image back onto a card/SSD – completely overwrites any existing data on the target.
`rsync -avh <source> <target>`	Compares two folders and only transfers what's new or changed – much faster than copying everything again from scratch. `a` also preserves permissions and timestamps, `v` lists every file individually, and `h` makes sizes readable (MB/GB instead of bytes).

Users & Permissions

Command	What it does
`sudo adduser <name>`	Interactively creates a new user with a home directory, asking for a password and optional details.
`sudo deluser <name>`	Removes a user account; the home directory is kept by default (use `--remove-home` to delete it too).
`sudo usermod -aG sudo <name>`	Adds the user to the sudo group without removing them from any group they're already in (that's what the `-a` flag does) – without it, all other group memberships would be lost.
`groups <name>`	Shows every group a user belongs to – e.g. to check whether they're in `gpio` or `i2c`.
`who`	Lists every currently logged-in user with terminal and login time, even across multiple simultaneous SSH sessions.
`sudo -i`	Opens a session where you work permanently as administrator (root), instead of prefixing every single command with `sudo` – until you leave it with `exit`. Use with care, there's no safety net here anymore.

Keyboard Shortcuts & Terminal Tricks

Shortcut	What it does
`Ctrl + C`	Immediately cancels whatever command is currently running – the terminal's emergency stop button when something hangs or you change your mind.
`Ctrl + Z`	Pauses the process and sends it to the background – bring it back with `fg`, or keep it running in the background with `bg`.
`Ctrl + R`	Starts an interactive reverse search through command history – typing filters live, Enter runs it.
`Tab`	Auto-completes commands, filenames, and paths – press twice to show all options when there's ambiguity.
`history`	Shows a numbered list of recently run commands – use `!123` to rerun command number 123 directly.
`!!`	Runs the last command again exactly as typed, without retyping it.
`sudo !!`	Repeats the last command with sudo prepended – handy when a command just failed due to missing permissions.
`clear`	Clears the visible terminal output, though the scroll history remains accessible via the scrollbar.
`exit`	Ends the current shell session – over an SSH connection, this also disconnects from the Pi.

If you regularly run new scripts on your Pi, also check out our autostart guide – so programs launch automatically after every reboot without typing a command by hand.

How do I find my Raspberry Pi's IP address from the terminal?

The fastest way is `hostname -I` directly on the Pi – it shows all assigned IP addresses at a glance. Alternatively, `ip a` gives a more detailed overview of all network interfaces.

Why doesn't the gpio readall command work anymore?

The `gpio` tool from the WiringPi library was removed from the official Raspberry Pi OS repositories because WiringPi hasn't been actively maintained for a long time. `raspi-gpio` is the replacement: `raspi-gpio get` shows the state of all pins, `raspi-gpio funcs` shows the available alternate functions.

How do I check if my Raspberry Pi is being throttled due to under-voltage?

Check with `vcgencmd get_throttled`. `0x0` means everything's fine. Any other value shows that the Pi has been throttled due to low power at some point, either right now or since the last boot – usually a sign of a too-weak power supply or a bad cable.

What's the difference between apt upgrade and apt full-upgrade?

`apt upgrade` only updates already installed packages and won't install new ones or remove existing ones. `apt full-upgrade` may also remove or install packages if that's needed to resolve dependencies during larger updates.

How do I create a full backup of my SD card from the command line?

`sudo dd if=/dev/sdX of=backup.img bs=4M status=progress` creates a 1:1 image of the entire card. Important: identify the correct drive with `lsblk` first, since mixing up `if=` and `of=` will overwrite the wrong target.

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